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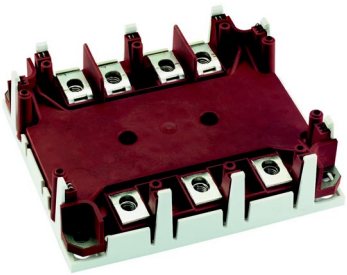
Absolute Maximum Ratings		Values	Units
Symbol	Conditions [1]		
V_{CES}		1200	V
V_{CGR}	R_{GE} = 20 kΩ	1200	V
I_C	T_{HS} = 25/70 °C	235 / 180	A
I_{CM}	T_{HS} = 25/70 °C; t_p = 1 ms	470 / 360	A
V_{GES}		± 20	V
P_{tot}	per IGBT, T_{HS} = 25 °C	625	W
T_j, (T_{stg})		– 40 ... +150 (125)	°C
T_{cop}	max. case operating temperature	125	°C
V_{isol}	AC, 1 min.	2500	V
humidity	IEC-EN 60721-3-3		
climate	IEC 68 T.1	40/125/56	
Inverse Diode			
$I_F = -I_C$	T_{HS} = 25/70 °C	230 / 180	A
$I_{FM} = -I_{CM}$	T_{HS} = 25/70 °C; t_p = 1 ms	460 / 360	A
I_{FSM}	t_p = 10 ms; sin.; T_j = 150 °C	2200	A
I^2t	t_p = 10 ms; T_j = 150 °C	24 200	A^2s

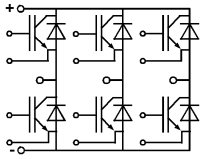
Characteristics		min.	typ.	max.	Units
Symbol	Conditions [1]				
$V_{(BR)CES}$	V_{GE} = 0, I_C = 1 mA	≥ V_{CES}	–	–	V
$V_{GE(th)}$	V_{GE} = V_{CE}, I_C = 4 mA	4,5	5,5	6,5	V
I_{CES}	V_{GE} = 0				
	V_{CE} = V_{CES}, T_j = 125 °C	–	15	–	mA
I_{GES}	V_{GE} = 20 V, V_{CE} = 0	–	–	500	nA
V_{CEsat} [4]	I_C = 200 A; V_{GE} = 15 V; T_j = 25 °C	–	2,0	2,3	V
C_{ies}	V_{GE} = 0	–	18		nF
C_{oes}	V_{CE} = 25 V	–	4,3		nF
C_{res}	f = 1 MHz	–	3,6	–	nF
L_{CE}		–	–	20	nH
$R_{CC'+EE'}$	resistance, terminal-chip; T_{HS} = 25 °C	–	1,15	–	mΩ
$t_{d(on)}$	V_{CC} = 600 V	–	150	–	ns
t_r	V_{GE} = +15 V / –15 V [3]	–	45	–	ns
$t_{d(off)}$	I_C = 200 A, ind. load	–	700	–	ns
t_f	R_{Gon} = R_{Goff} = 5 Ω	–	50	–	ns
E_{on}	T_j = 125 °C	–	21	–	mJ
E_{off}		–	20	–	mJ
Inverse Diode [8]					
$V_F = V_{EC}$	I_F = 200 A; V_{GE} = 0 V; T_j = 25 (125) °C	–	2,3 (2,1)	2,6	V
$V_F = V_{EC}$	I_F = 100 A; T_j = 25 (125) °C	–	1,8 (1,6)		V
V_{TO}	T_j = 125 °C	–	1,1		V
r_T	T_j = 125 °C	–	5		mΩ
I_{RRM}	I_F = 200 A; T_j = 25 (125) °C [2]	–	TBD	–	A
Q_{rr}	I_F = 200 A; T_j = 25 (125) °C [2]	–	TBD	–	µC
Thermal Characteristics [5]					
R_{thjh}	per IGBT	–	–	0,20	°C/W
R_{thjhD}	per diode	–	–	0,285	°C/W
R'_{thjc} [6]	per IGBT	–	–	0,071	°C/W
R'_{thjcD} [6]	per diode	–	–	0,105	°C/W
Temperature Sensor					
R_{TS}	T = 25 °C / 100 °C	–	1,0 / 1,67	–	kΩ
tolerance	T = 25 °C / 100 °C	–	3,0 / 2,0	–	%

SKiM® 4 IGBT Modules

SKiM 250 GD 128 D

Preliminary Data





GD

Features

- N channel, homogeneous planar IGBT Silicon structure with n+ buffer layer in SPT (soft punch through) technology
- Low inductance case
- Fast & soft inverse CAL diodes [8]
- Isolated by DCB (Direct Copper Bonded) ceramic plate
- Pressure contact technology for thermal contacts
- Spring contact system to attach driver PCB to the control terminals
- Integrated temperature sensor

Typical Applications

- Switched mode power supplies
- Three phase inverters for AC motor speed control
- Switching (not for linear use)

[1] T_{HS} = 25 °C, unless otherwise specified
[2] TBD
[3] Use V_{GEoff} = – 5 ... – 15 V
[4] Measured at chip level
[5] See mounting instructions
[6] Corresponding value. This value cannot be measured. It is only given for comparison.
[8] CAL = Controlled Axial Lifetime Technology

This datasheet has been downloaded from http://www.digchip.com at this page

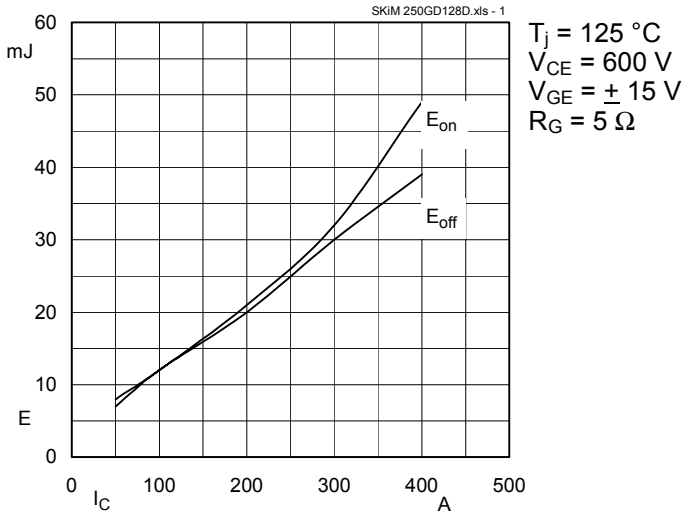


Fig. 1 Turn-on /-off energy = f (I_C)

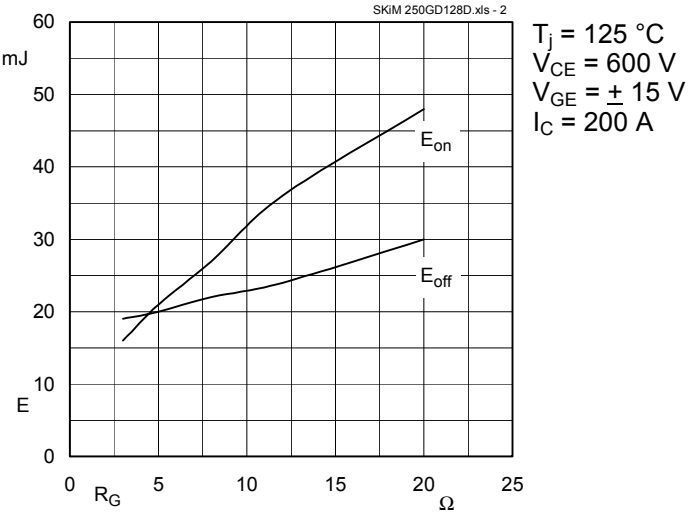


Fig. 2 Turn-on /-off energy = f (R_G)

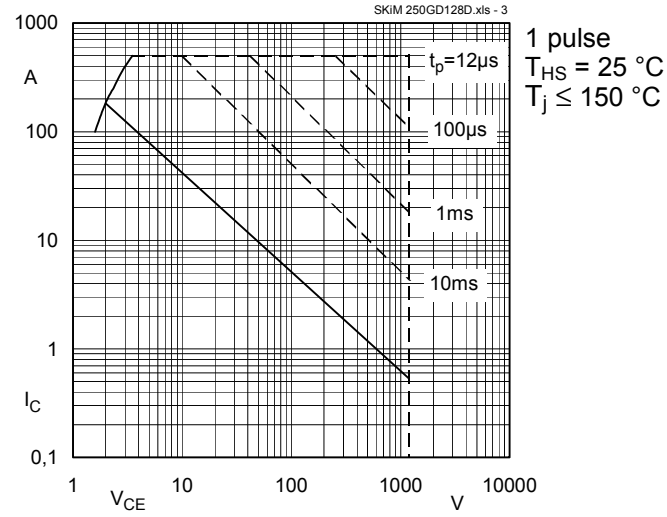


Fig. 3 Maximum safe operating area (SOA) I_C = f (V_{CE})

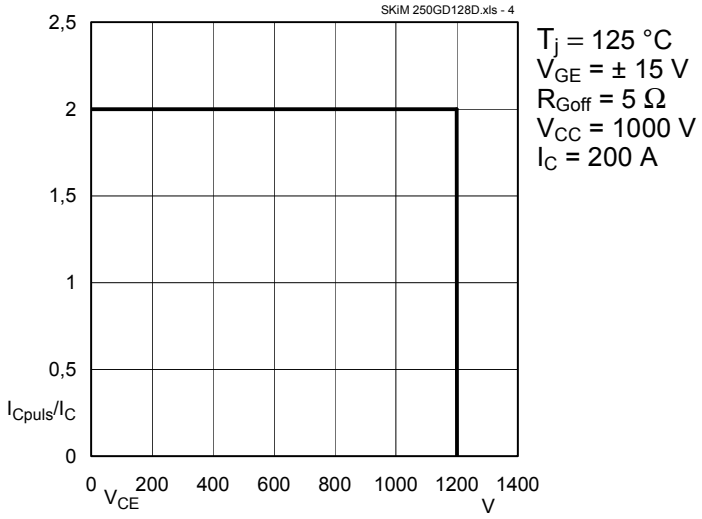


Fig. 4 Turn-off safe operating area (RBSOA)

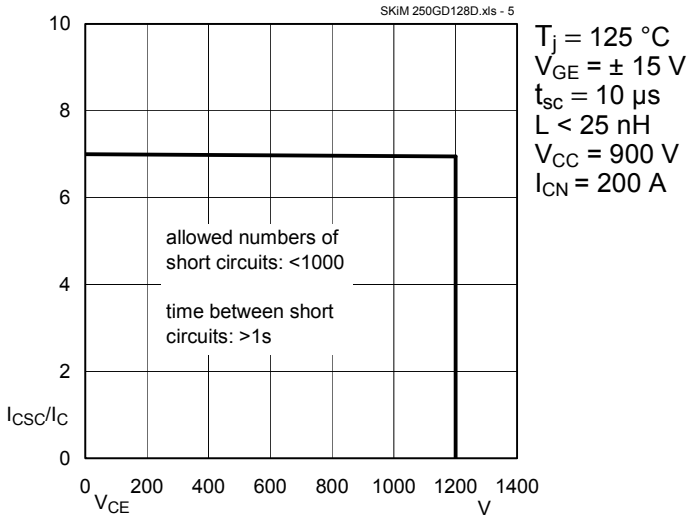


Fig. 5 Safe operating area at short circuit I_C = f (V_{CE})

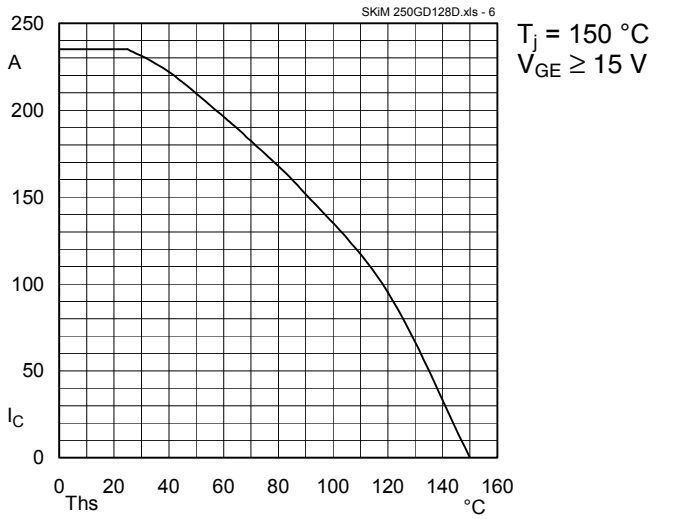


Fig. 6 Rated current vs. temperature I_C = f (T_{HS})

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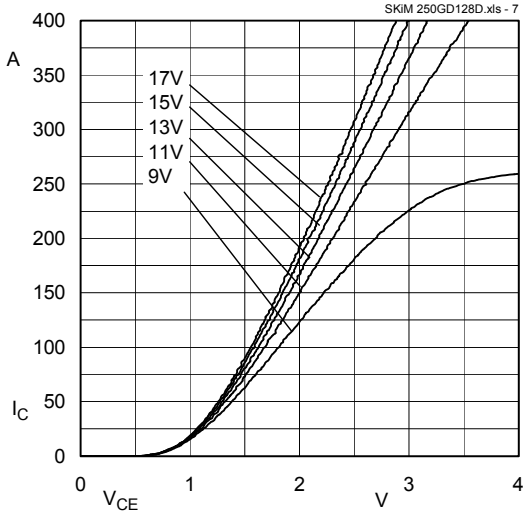


Fig. 7 Typ. output characteristic, t_p = 80 µs; 25 °C

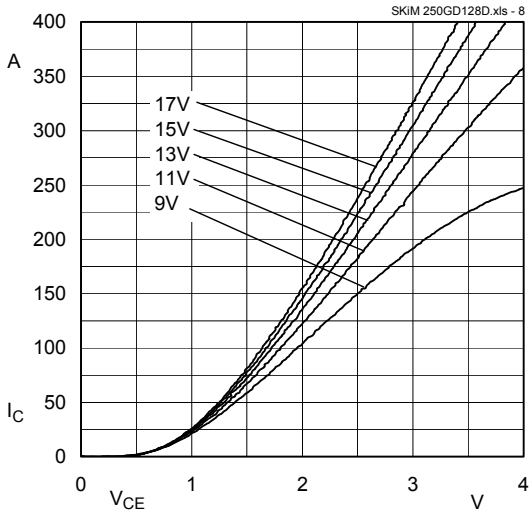


Fig. 8 Typ. output characteristic, t_p = 80 µs; 125 °C

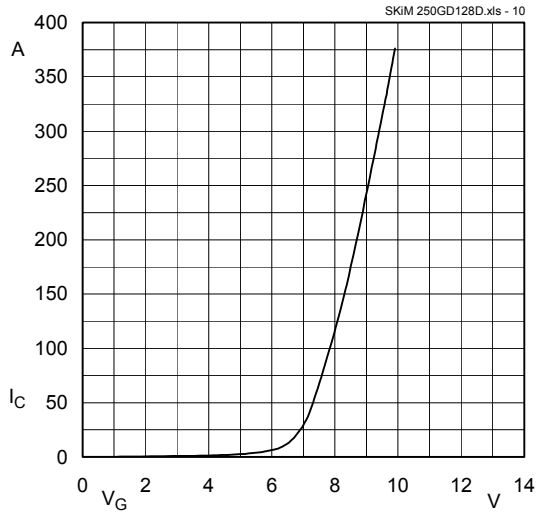
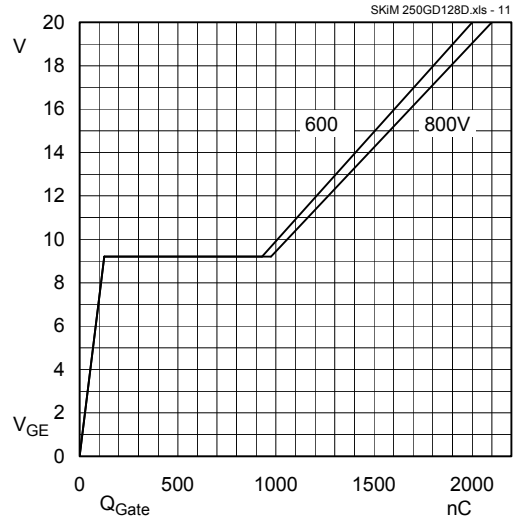


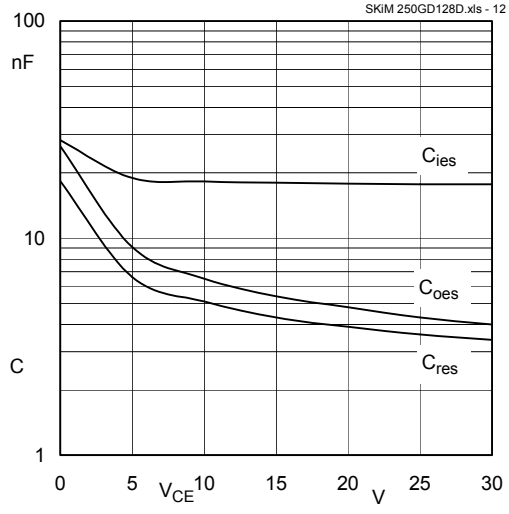
Fig. 9 Saturation characteristic (IGBT)
Calculation elements and equations

Fig. 10 Typ. transfer characteristic, t_p = 80 µs; V_{CE} = 20 V



I_{Cpuls} = 200 A

Fig. 11 Typ. gate charge characteristic



V_{GE} = 0 V
f = 1 MHz

Fig. 12 Typ. capacitances vs. V_{CE}

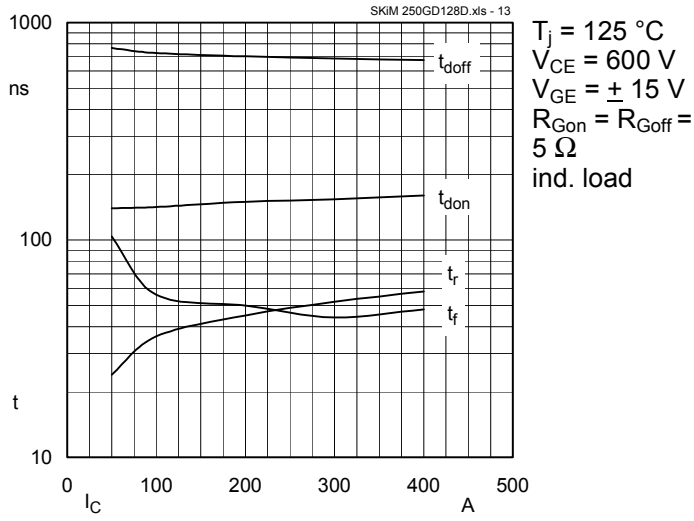


Fig. 13 Typ. switch times vs. I_C

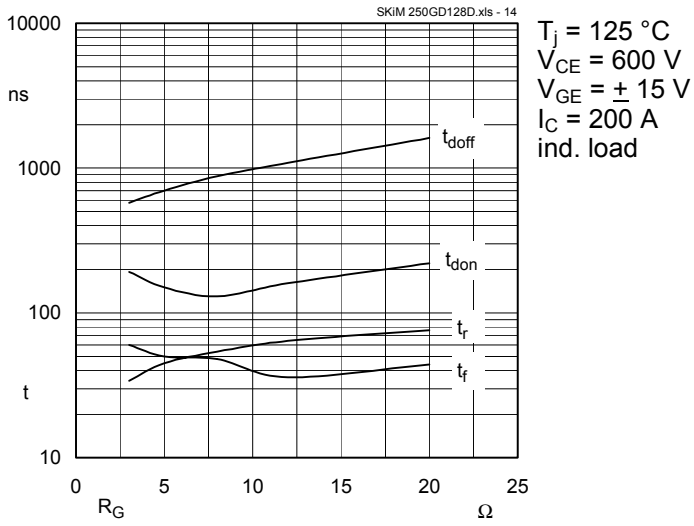


Fig. 14 Typ. switch times vs. gate resistor R_G

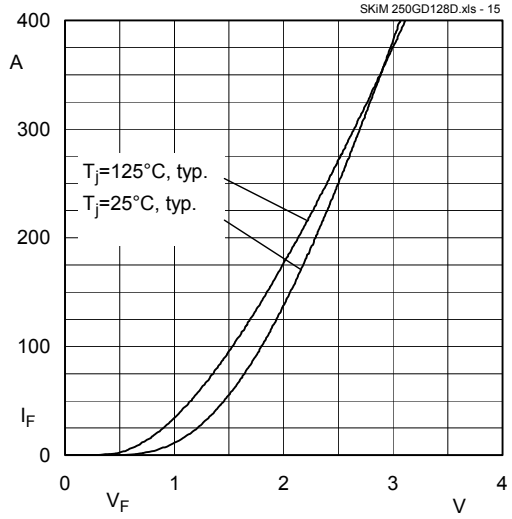


Fig. 15 Typ. CAL diode forward characteristic

Fig. 16 Diode turn-off energy dissipation per pulse

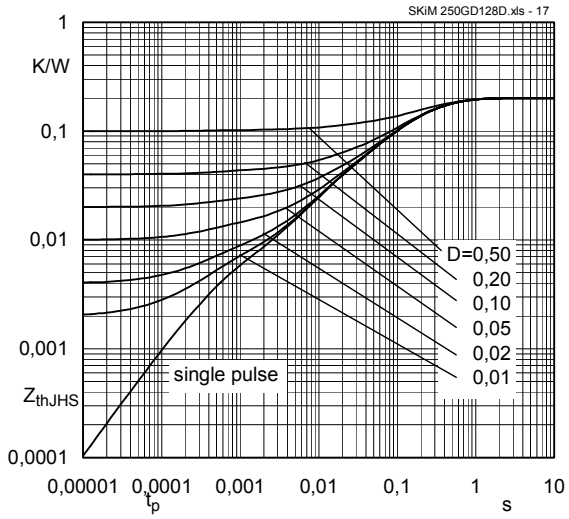


Fig. 17 Transient thermal impedance of IGBT
Z_{thJHS} = f (t_p); D = t_p / t_c = t_p · f

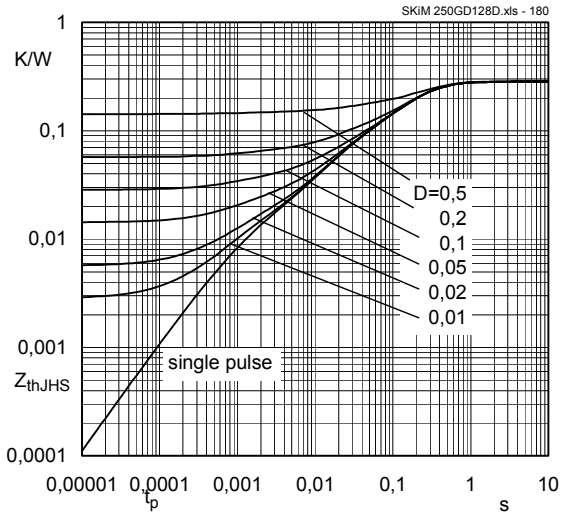
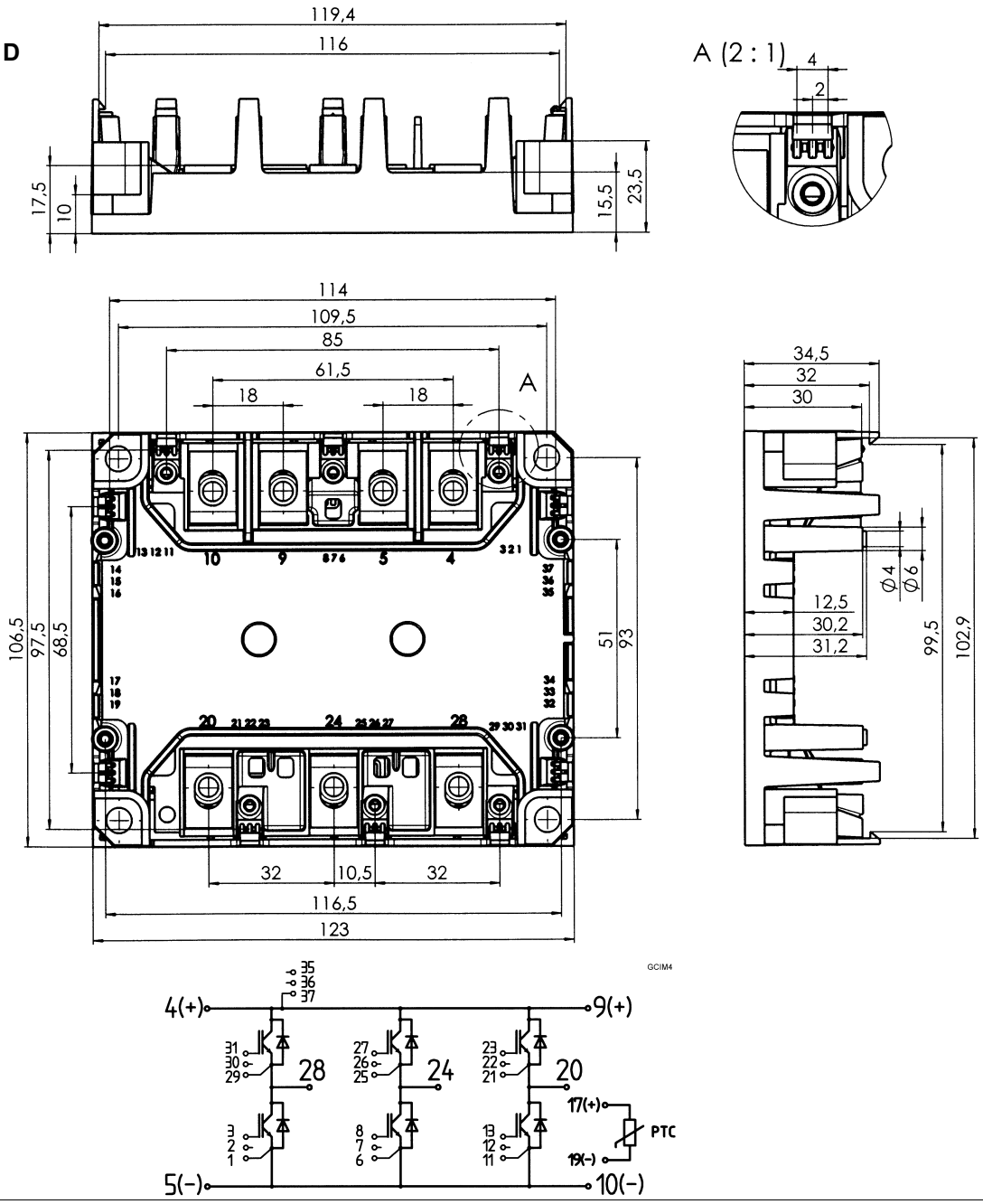


Fig. 18 Transient thermal impedance of inverse CAL diodes Z_{thJHS} = f (t_p); D = t_p / t_c = t_p · f

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SKiM 4
SKiM 250 GD 128 D



Dimensions in mm

Case outline and circuit diagram

Mechanical Data						
Symbol	Conditions		Values			Units
			min.	typ.	max.	
M_1	to heatsink, SI Units	(M5)	2	–	3	Nm
	to heatsink, US Units		18	–	26	lb.in.
M_2	for terminals, SI Units	(M6)	4	–	5	Nm
	for terminals, US Units		35	–	44	lb.in.
a			–	–	5x9,81	m/s^2
w			–	–	310	g

This is an electrostatic discharge sensitive device (ESDS).
Please observe the international standard IEC 747-1, Chapter IX.

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